Problem 1

During the first half of the total travel time, a car moves at a speed of $v_1 = 54\,\mathrm{km/h}$. During the second half of the travel time, the car moves at a speed of $v_2 = 36\,\mathrm{km/h}$. What is the average speed of the car?

Solution

To find the average speed when the time intervals are equal, we use the formula:

$$v_{\mathrm{avg}} = \frac{2v_1 v_2}{v_1 + v_2}.$$

Given:

$$v_1 = 54\,\mathrm{km/h}, \quad v_2 = 36\,\mathrm{km/h},$$

Substituting the values:

$$v_{\mathrm{avg}} = \frac{2v_1 v_2}{v_1 + v_2} = \frac{2 \cdot 54 \cdot 36}{54 + 36}.$$

Simplify:

$$v_{\mathrm{avg}} = \frac{2 \cdot 1944}{90} = \frac{3888}{90} = 43.2\,\mathrm{km/h}.$$

Thus, the average speed of the car is:

$$v_{\mathrm{avg}} = 43.2\,\mathrm{km/h}.$$